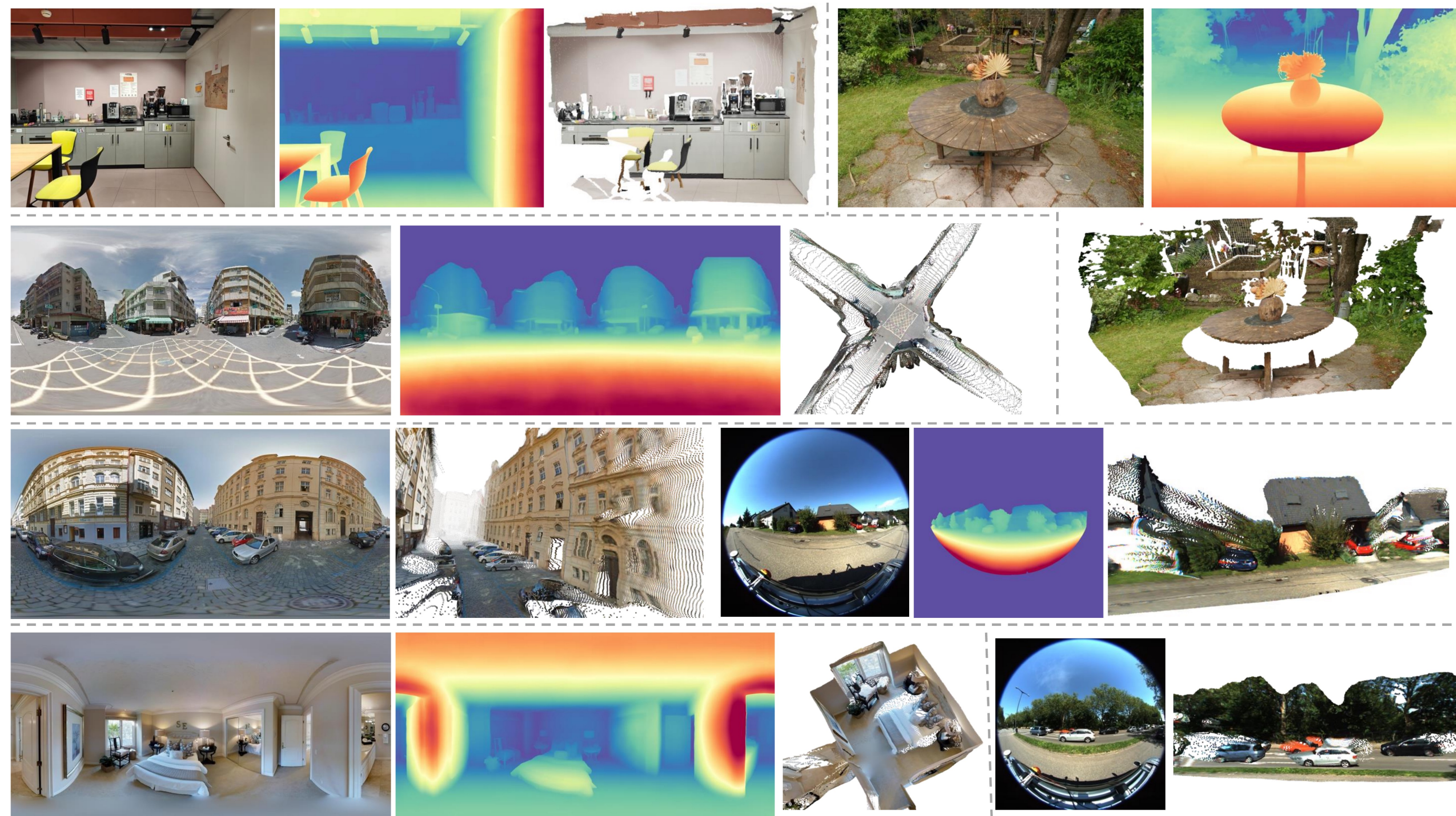


1. Motivation



Metric point clouds from a single image — pinhole, fisheye, 360° — with intrinsics and sparse depth optional.

☹️ **Fragmented today:** a separate model per projection; priors cannot be toggled.

😊 **OmniPoint:** one network, any camera, priors optional.

The only method covering every output, camera and prior

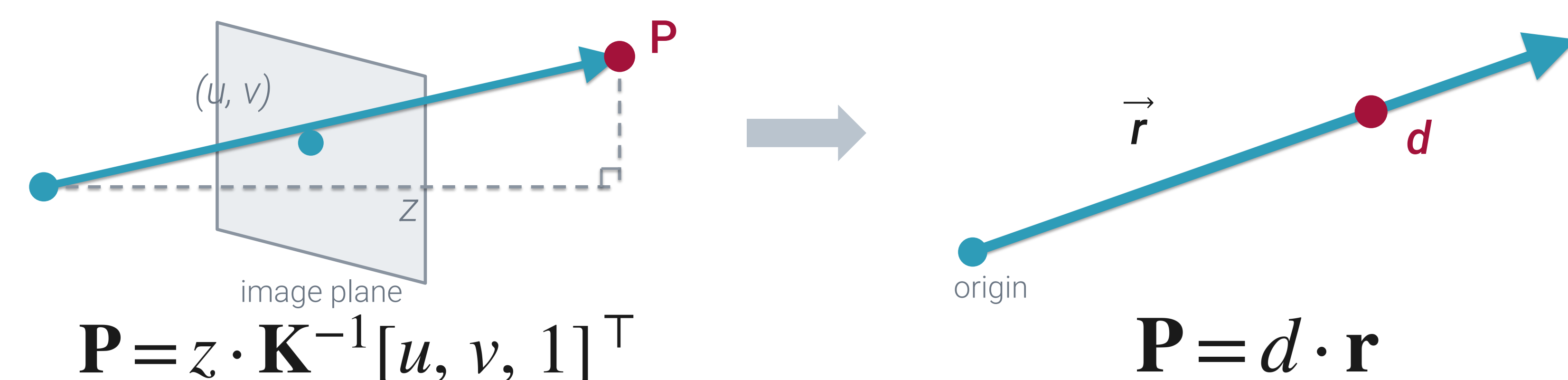
Capability	DA v2	MoGe V2	Depth Pro	UniK3D	DA ²	PromptDA	OmniPoint
Affine-inv. depth	✓	✓	✓	✓	—	✓	✓
Affine-inv. points	—	✓	✓	✓	—	—	✓
Metric points	—	✓	✓	✓	—	—	✓
Pinhole	✓	✓	✓	✓	—	✓	✓
Fisheye	—	—	—	✓	—	—	✓
360° panorama	—	—	—	✓	✓	—	✓
Intrinsics prior	—	—	—	✓	—	—	✓
Sparse-depth prior	—	—	—	—	—	✓	✓

2. Camera-Agnostic Ray + Distance

☹️ **Planar depth z:** needs a linear pixel→ray map; undefined past 180° FoV.

☹️ **Raw (x, y, z):** entangles projection with scene structure.

😊 **Ray × distance:** r absorbs the camera, d carries the scene.



Decoupled training objective

$$\mathcal{L}_{\text{ray}} = \sum_i \|\hat{\mathbf{r}}_i - \mathbf{r}_i\|_1 \quad \mathcal{L}_{\text{point}} = \sum_i \|s^* \hat{\mathbf{d}}_i \cdot \mathbf{r}_i - d_i \cdot \mathbf{r}_i\|_1$$

Distance is supervised along the **ground-truth ray** — ray errors never pollute it.

Representation	S.FoV	L.FoV	360°	Point loss	Rel↓
Ray + D (ours)	4.04	6.37	5.79	P + GT ray (ours)	5.55
XYZ point map	4.28	6.74	6.28	naive P	5.65
				disjoint D + Ray	7.88

3. OmniPoint Pipeline

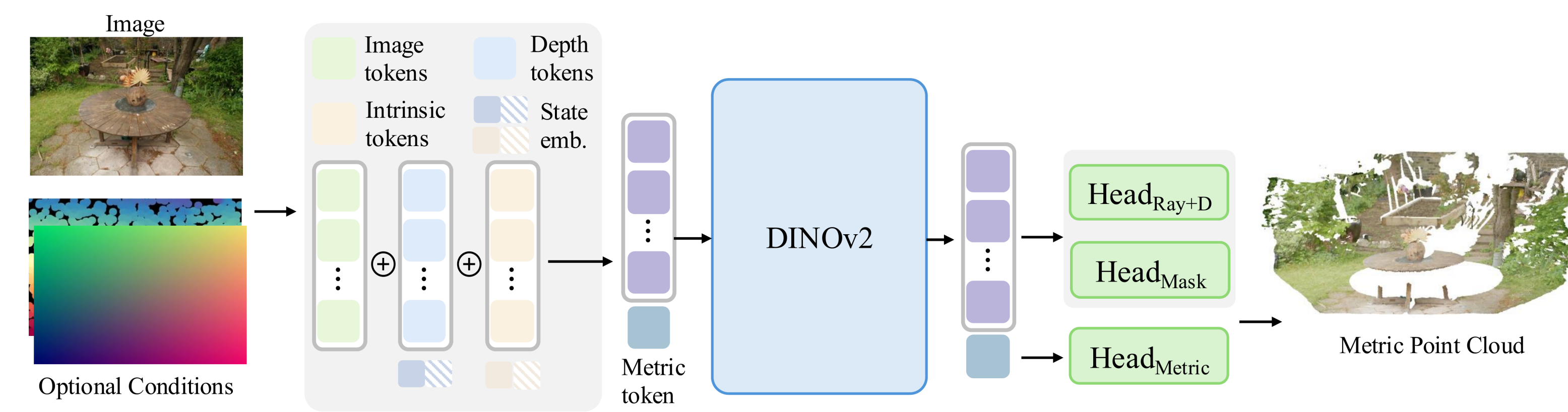


Image tokens + optional intrinsic / sparse-depth tokens → one DINOv2 ViT-L → DPT heads for ray + distance, sky mask, metric scale.

Optional priors — without destabilising the backbone

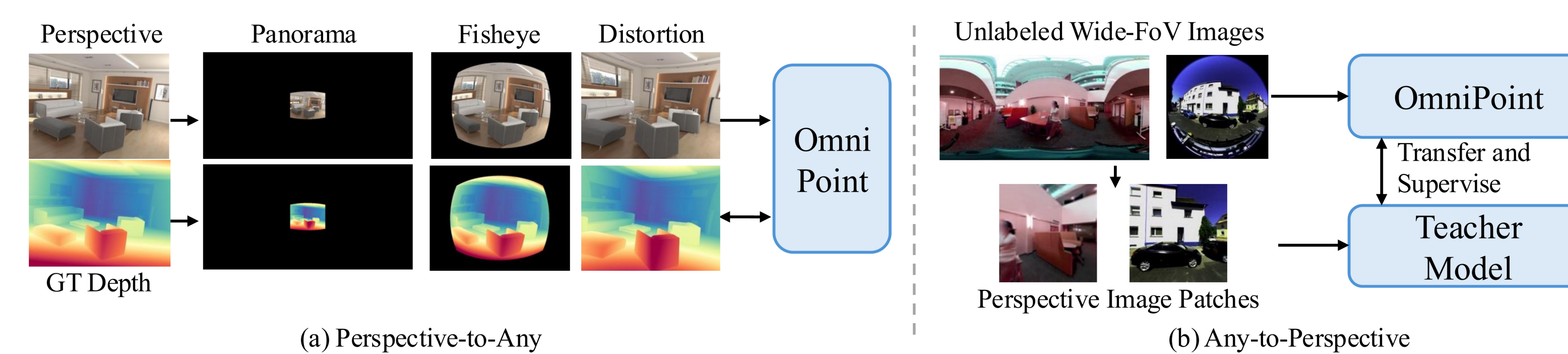
- ① **Input-state embeddings:** tokens tell the ViT which priors are active.
- ② **Gaussian smoothing:** sparse points become a dense, consistent guide.

$$w(\Delta u, \Delta v) = \exp\left(-\frac{\Delta u^2 + \Delta v^2}{2\sigma^2}\right)$$

Conditioning design (sparse depth)	Rel↓	δ ₁ ↑
Ours (full)	3.21	98.2
w/o Gaussian smoothing	3.80	97.5
w/o input-state embedding	3.38	97.7

29 labelled + 2 unlabelled datasets · 72 A100s.

4. Bidirectional Data Augmentation

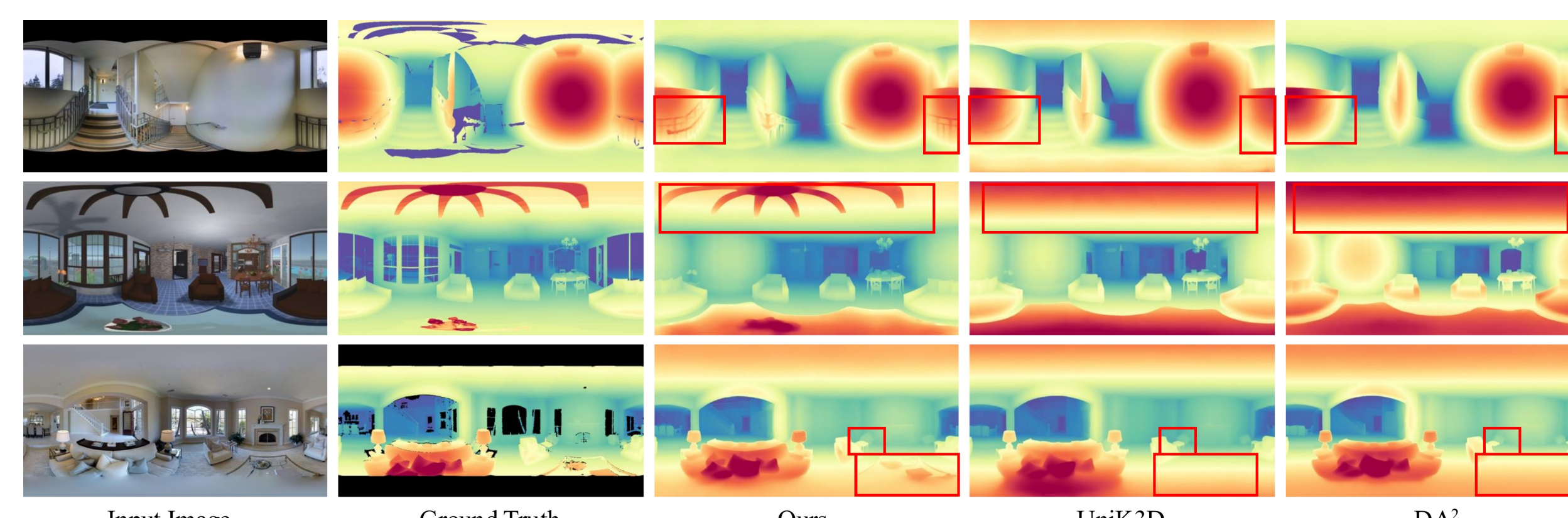


Non-pinhole ground truth is scarce — so we bridge labelled perspective and unlabelled 360° data in 3D.

→ **Perspective-to-Any:** re-project labelled pinhole data into virtual fisheye and 360° cameras.

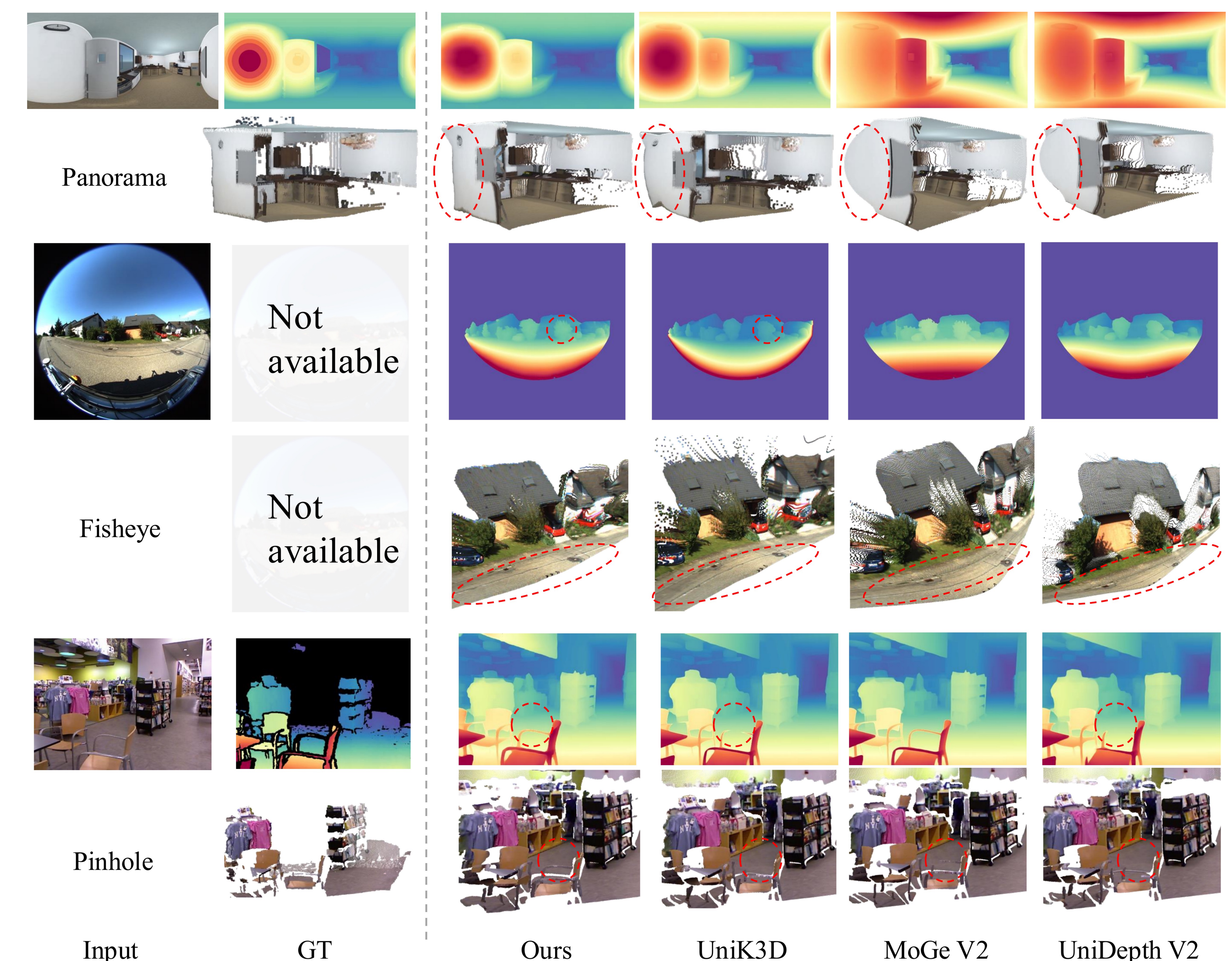
← **Any-to-Perspective:** pseudo-label virtual pinhole crops of unlabelled wide-FoV images.

P2A	A2P	360° Rel↓	360° δ ₁ ↑
—	—	8.45	89.8
✓	—	6.21	94.5
—	✓	8.21	90.9
✓	✓	5.79	95.1



Sharper panorama depth than the specialist DA².

5. Qualitative Results



Vertical walls, straight roads, sharp boundaries — exactly where UniK3D's spherical-harmonic ray field smooths detail away.

6. Quantitative Results

6.66

fisheye point Rel↓
UniK3D 11.5

5.79

360° depth Rel↓
DA² 6.65

83.5

metric δ₁ mean
best image-only

98.0

metric δ₁ mean
with sparse depth

Zero-shot relative geometry — Rel↓ (%)

Method	S.FoV		L.FoV		360° Images	
	Depth	Point	Depth	Point	Depth	Point
UniDepth V2	4.23	6.06	7.81	18.4	19.3	102.8
Depth Pro	5.28	7.49	26.6	36.2	—	—
MoGe V1	4.05	5.25	9.48	28.1	22.3	102.6
MoGe V2	3.98	5.59	12.0	23.4	25.1	102.8
DA3	4.78	6.33	12.8	29.9	25.5	101.5
UniK3D	4.14	5.61	8.77	11.5	10.4	11.4
DA²	—	—	—	—	6.65	7.30
OmniPoint (ours)	4.04	5.55	6.37	6.66	5.79	5.90

Zero-shot metric depth — δ₁↑ (%)

Method	NYUv2	KITTI	ETH3D	iBims-1	DIODE	HAMMER	Mean
UniDepth V2	92.8	95.4	69.5	93.2	51.8	46.8	74.4
Depth Pro	91.9	38.3	32.8	81.5	37.7	63.0	57.5
MoGe V2	96.1	62.9	90.8	83.0	66.4	65.6	77.5
UniK3D	94.4	93.6	83.7	92.8	73.0	58.3	82.6
OmniPoint (ours)	86.2	88.0	91.8	87.7	73.3	73.8	83.5
+ intrinsics	88.3	90.7	93.0	88.3	75.2	74.5	85.0
+ sparse depth	98.5	98.4	94.1	99.0	98.4	99.3	98.0